PRAMODHA M

Assistant Professor | Deep Learning Researcher
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Google Scholar | LinkedIn | GitHub | ResearchGate

PROFESSIONAL SUMMARY

Assistant Professor and Ph.D. research scholar in Computer Science with 3+ years of experience in machine learning research and software testing. Research focus on explainable deep learning for medical imaging and agriculture (IoMT/IoAT), including few-shot learning and image segmentation. Teaches Design and Analysis of Algorithms. Best Paper Award at IOCA 2021, with six peer-reviewed publications.

EDUCATION

Ph.D. in Computer Science (in progress) 2025 – Present
University of Mysore, Karnataka, India
Research area: deep learning for medical and agricultural imaging (IoMT/IoAT). Thesis: Edge-IoT Based Plant Pathology Framework Using Hybrid Deep Learning Approaches

Master of Computer Applications (MCA) 2016 – 2019
Department of Studies in Computer Science, Manasagangotri, University of Mysore, Karnataka, India

Bachelor of Science (B.Sc.) 2011 – 2015
Yuvaraja's College, University of Mysore, Karnataka, India

PROFESSIONAL EXPERIENCE

Assistant Professor 2025 – Present
Department of Computer Science and Engineering, School of Engineering, University of Mysore
- Teach Design and Analysis of Algorithms using real-life analogies, step-by-step traced examples, and pseudocode before code.
- Lead research on explainable and few-shot deep learning models for blood cell analysis, plant disease recognition, and agricultural image datasets.

Project Associate Oct 2023 – Mar 2025
CSIR – Central Food Technological Research Institute (CFTRI), Mysuru
- Led an end-to-end machine learning pipeline for near-infrared (NIR) spectroscopy data: data curation, preprocessing, feature engineering, dimensionality reduction, and predictive model optimisation.
- Worked with chemists and domain scientists to build domain knowledge into model design, improving model accuracy.
- Documented processes and results so that experiments are reproducible.

Software Test Engineer Feb 2022 – Oct 2023
Accenture, Bengaluru
- Performed manual and automated testing on web and mobile platforms (Android, iOS, Windows, Mac).
- Executed regression, performance, and user acceptance testing in Agile project teams.
- Improved QA delivery by streamlining bug triage and working directly with developers on faster fixes.
- Mentored junior testers on manual testing practices and methodology.

SELECTED RESEARCH PROJECTS

- **Eff-UNet3+:** Explainable segmentation of white blood cells using an EfficientNet-based UNet3+ with explainability methods (in progress).
- **Few-shot blood cell classification:** Prototypical Networks for 13-class white blood cell classification from few labelled images, with a MobileNetV4 backbone (in progress).
- **Few-shot leaf disease classification:** Prototypical Network with a Swin-Tiny backbone for plant leaf disease recognition (in progress).
- **Mulberry leaf image dataset:** Image collection pipeline across lab, greenhouse, and field conditions for the university silkworm-feed plantation.
- **COVID-19 detection (2019–2020):** CNN-based classifier separating COVID-19 cases from chest X-ray and CT images; basis of the IOCA 2021 Best Paper.
- **Human blood cell classification (2020):** Deep learning framework for classifying peripheral blood cells from microscopy images.

PUBLICATIONS

[1] C. Chola, **M. Pramodha**, J. Hanumanthappa, B. Anjali, A. Y. Muaad, and A. Tiwari, "XAI-driven fine-tuned EfficientNetV2 model for white blood cell classification," in *Artificial Intelligence and Knowledge Processing (AIKP 2025)*, H. Kannan et al., Eds., CCIS, vol. 2804. Cham: Springer, 2026. doi: 10.1007/978-3-032-14706-6_4.

[2] B. Ashoka S., **M. Pramodha**, A. Y. Muaad, R. Nyange, A. Anusha, and N. Shilpa G., "Explainable AI based framework for banana disease detection," in *Proc. 5th Int. Conf. Innovative Trends in Information Technology (ICITIIT)*, IEEE, 2024. doi: 10.1109/ICITIIT61487.2024.10580364.

[3] **M. Pramodha**, S. Ansith, J. V. B. Benifa, M. Al-Sarem, J. Hanumanthappa, A. A. Bini, E. Ndagijimana, F. Saeed, M. B. B. Heyat, A. Alqarafi, A. Y. Muaad, and C. Chola, "A deep learning model for human blood cells classification," *Lecture Notes on Data Engineering and Communications Technologies*, 2023. doi: 10.1007/978-3-031-36258-3_36.

[4] A. Y. Muaad, G. Hemanth Kumar, J. Hanumanthappa, J. V. Bibal Benifa, M. Naveen Mourya, C. Chola, **M. Pramodha**, and R. Bhairava, "An effective approach for Arabic document classification using machine learning," *Global Transitions Proceedings*, 2022. doi: 10.1016/j.gltp.2022.03.003.

[5] J. Hanumanthappa, A. Y. Muaad, J. V. Bibal Benifa, C. Chola, V. Hiremath, and **M. Pramodha**, "IoT-based smart diagnosis system for health care," in *Sustainable Communication Networks and Application*. Singapore: Springer, 2021. doi: 10.1007/978-981-16-6605-6_1.

[6] **M. Pramodha**, A. Y. Muaad, J. V. Bibal Benifa, J. Hanumanthappa, C. Chola, and M. A. Al-antari, "A hybrid deep learning approach for Covid-19 diagnosis via CT and X-ray medical images," in *Proc. 1st Int. Electronic Conf. on Algorithms (IOCA 2021)*, MDPI, 2021. doi: 10.3390/IOCA2021-10909. **Best Paper Award.**

HONORS AND AWARDS

Best Paper Award 2021
The 1st International Electronic Conference on Algorithms (IOCA 2021), MDPI

TECHNICAL SKILLS

Programming	Python, C/C++, SQL/MySQL, PHP
Machine learning	PyTorch, TensorFlow, NumPy, Pandas
Research areas	Medical image analysis, explainable AI, few-shot learning, image segmentation, NIR spectroscopy modelling, IoMT/IoAT
Software testing	Manual and automation testing, regression, performance and UAT testing, bug triage, Agile
Teaching	Design and Analysis of Algorithms; curriculum and example-driven lesson design